Simulation Methods for Finance

MATH70113 — Group Coursework Report

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Contents

1 Introduction **2**

2 Part I: Pathwise Sensitivity for a Down-and-Out Barrier Option **2**
2.1 Model and Option Specification . . . 2
2.2 Pathwise Sensitivity: Mathematical Derivation . . . 2
2.2.1 Pathwise Derivatives of S_t . . . 3
2.2.2 Complete Delta Estimator . . . 3
2.2.3 Complete Vega Estimator . . . 3
2.3 Brownian Bridge Barrier Correction . . . 4
2.3.1 Motivation and Order of Bias . . . 4
2.3.2 Crossing Probability via the Brownian Bridge . . . 4
2.3.3 Survival Probability . . . 4
2.4 Implementation Details . . . 4
2.4.1 Choice of M . . . 5
2.5 Numerical Results . . . 5
2.5.1 Reference Values . . . 5
2.5.2 Convergence Analysis . . . 6

3 Part II: Number of Paths vs Number of Basis Functions **7**
3.1 Introduction and Motivation . . . 7
3.2 Mathematical Framework . . . 7
3.2.1 The Optimal Stopping Problem . . . 7
3.2.2 Quasi-Regression Approximation . . . 8
3.2.3 Mean Squared Error Bounds . . . 8
3.2.4 Hermite Basis for Brownian Motion . . . 8
3.2.5 Power Basis for Geometric Brownian Motion . . . 8
3.3 Main Theoretical Results . . . 9
3.3.1 The Variance Explosion Mechanism . . . 9
3.3.2 Theorem 1 (Normal Setting) . . . 9
3.3.3 Theorem 2 (Lognormal Setting) . . . 9
3.3.4 Theorem 3 (Multiperiod Extension) . . . 10
3.4 Numerical Reproduction . . . 10
3.4.1 Setup: Brownian Case ($t_1 = 1$, $t_2 = 2$, $\rho = 2$) . . . 10
3.4.2 Reproduced Table: Brownian Case . . . 10
3.4.3 Lognormal Case ($t_1 = 1$, $t_2 = 2$) . . . 11
3.5 Analysis: The Phase Transition . . . 12
3.5.1 Orthogonality and the Scope of the Analysis . . . 13
3.6 Critical Assessment . . . 13
3.6.1 Strengths . . . 13
3.6.2 Limitations and Extensions . . . 13
3.7 From Theory to Practice: American Put Pricing Experiment . . . 14

4 Conclusion **15**

Task Division **16**

1 Introduction

Monte Carlo methods occupy a central role in computational finance, offering flexible and scalable tools for pricing complex derivatives and computing their sensitivities when analytical solutions are unavailable. This report investigates two complementary aspects of Monte Carlo simulation in finance.

Part I addresses the estimation of Delta and Vega for a down-and-out barrier call option under the Black–Scholes framework. We implement the *pathwise sensitivity* method [Glasserman, 2004] and incorporate a Brownian bridge correction to account for barrier crossings between discretisation points, achieving $\mathcal{O}(\Delta t)$ weak convergence in theory rather than the naive $\mathcal{O}(\sqrt{\Delta t})$, a rate consistent with but not directly provable from our simulations: the BB residual bias falls below the Monte Carlo noise floor at all practical sample sizes (Figure 3). The derivation pays careful attention to the product-rule contributions arising from the dependence of the survival probability on the model parameters; we show numerically (Remark 2.1) that this correction accounts for 74% of $\hat{\Delta}$ near the barrier.

Part II studies the regression-based Monte Carlo approach to American option pricing introduced by Longstaff and Schwartz [2001], analysed theoretically by Glasserman and Yu [2004]. We present the full mathematical framework, reproduce the key numerical experiment of the paper (Table 3), extend it to the lognormal case, and demonstrate the practical implications of the theoretical bounds through a direct pricing experiment on an American put option.

2 Part I: Pathwise Sensitivity for a Down-and-Out Barrier Option

2.1 Model and Option Specification

The asset price S satisfies the Geometric Brownian Motion (GBM) under the risk-neutral measure:

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t, \qquad t \in [0,T], \quad S_0 = s_0, \tag{1}$$

with parameters $r = 0.05$, $\sigma = 0.5$, $T = 1$, $s_0 = 100$, $K = 110$, $B = 90$. The explicit solution is

$$S_t = s_0 \exp\Big(\big(r - \tfrac{\sigma^2}{2}\big)t + \sigma W_t\Big). \tag{2}$$

The discounted payoff of the down-and-out barrier call option is

$$V = \mathbb{E}\Big[e^{-rT}(S_T - K)^+\,\mathbf{1}_{\min_{t\in[0,T]} S_t > B}\Big]. \tag{3}$$

The indicator $\mathbf{1}_{\min S_t > B}$ nullifies the payoff if the asset ever touches the barrier B before maturity. Path-dependence makes direct simulation non-trivial: only a finite grid of times can be monitored.

2.2 Pathwise Sensitivity: Mathematical Derivation

The pathwise method [Glasserman, 2004, Section 7.2] exploits

$$\frac{\partial}{\partial\theta}\mathbb{E}[f(S;\theta)] = \mathbb{E}\left[\frac{\partial f}{\partial\theta}(S;\theta)\right],$$

valid whenever differentiation and expectation can be interchanged, and computes the derivative inside the Monte Carlo average. This interchange is justified here because: (i) the map $s_0 \mapsto (S_T - K)^+\,\mathbf{1}_{\min S_t > B}$ is differentiable $\mathbb{P}$-almost surely, the events $\{S_T = K\}$ and $\{S_T = B\}$ have probability zero under GBM, so the non-differentiabilities of the payoff and indicator are measure-zero sets; (ii) a dominating integrable derivative exists (see Glasserman [2004, Proposition 7.2.1]).

2.2.1 Pathwise Derivatives of S_t

From (2):

$$\frac{\partial S_t}{\partial s_0} = \frac{S_t}{s_0}, \qquad \frac{\partial S_t}{\partial \sigma} = S_t(W_t - \sigma t),$$
$$\frac{\partial \ln(S_t/B)}{\partial s_0} = \frac{1}{s_0}, \qquad \frac{\partial \ln(S_t/B)}{\partial \sigma} = W_t - \sigma t. \tag{4}$$

2.2.2 Complete Delta Estimator

Define the raw payoff $G = e^{-rT}(S_T - K)^+$ and let $\mathcal{S} = \prod_{i=0}^{M-1}(1 - p_i)$ denote the survival probability (defined precisely in Section 2.3). The product rule gives

$$\frac{\partial}{\partial s_0}\big[G \cdot \mathcal{S}\big] = \frac{\partial G}{\partial s_0} \cdot \mathcal{S} + G \cdot \frac{\partial \mathcal{S}}{\partial s_0}. \tag{5}$$

First term.

$$\frac{\partial G}{\partial s_0} = e^{-rT}\mathbf{1}_{S_T > K}\frac{S_T}{s_0}.$$

Second term. Since $\mathcal{S} = \prod_i(1 - p_i)$,

$$\frac{\partial \mathcal{S}}{\partial s_0} = \sum_{j=0}^{M-1} \frac{\partial(1 - p_j)}{\partial s_0} \prod_{i \neq j}(1 - p_i). \tag{6}$$

The crossing probability is $p_j = \exp\left(-\frac{2\ln(S_{t_j}/B)\ln(S_{t_{j+1}}/B)}{\sigma^2 \Delta t}\right)$, so using (4):

$$\frac{\partial p_j}{\partial s_0} = p_j \cdot \left(-\frac{2}{\sigma^2 \Delta t}\right) \cdot \frac{\ln(S_{t_j}/B) + \ln(S_{t_{j+1}}/B)}{s_0}, \tag{7}$$

and therefore

$$\frac{\partial(1 - p_j)}{\partial s_0} = \frac{2\,p_j}{\sigma^2 s_0 \Delta t}\big[\ln(S_{t_j}/B) + \ln(S_{t_{j+1}}/B)\big]. \tag{8}$$

Combining (5)–(8), the complete **Delta estimator** is

$$\boxed{\hat{\Delta} = e^{-rT}\left[\mathbf{1}_{S_T > K}\frac{S_T}{s_0} \cdot \mathcal{S} + (S_T - K)^+ \sum_{j=0}^{M-1} \frac{2\,p_j}{\sigma^2 s_0\,\Delta t}\Big[\ln(S_{t_j}/B) + \ln(S_{t_{j+1}}/B)\Big] \cdot \frac{\mathcal{S}}{1 - p_j}\right].} \tag{9}$$

2.2.3 Complete Vega Estimator

By the same product rule:

$$\hat{\mathcal{V}} = e^{-rT}\left[\mathbf{1}_{S_T > K}\, S_T(W_T - \sigma T) \cdot \mathcal{S} + (S_T - K)^+ \cdot \frac{\partial \mathcal{S}}{\partial \sigma}\right]. \tag{10}$$

Using (4):

$$\frac{\partial(1 - p_j)}{\partial \sigma} = \frac{2p_j}{\sigma^2 \Delta t}\left[(W_{t_j} - \sigma t_j)\ln\frac{S_{t_{j+1}}}{B} \;+\; \ln\frac{S_{t_j}}{B}\,(W_{t_{j+1}} - \sigma t_{j+1}) \;-\; \frac{2}{\sigma}\ln\frac{S_{t_j}}{B}\,\ln\frac{S_{t_{j+1}}}{B}\right]. \tag{11}$$

Substituting (11) into (10), and denoting by C_j^σ the bracket in (11), the complete **Vega estimator** is

$$\boxed{\hat{\mathcal{V}} = e^{-rT}\left[\mathbf{1}_{S_T > K}\, S_T(W_T - \sigma T) \cdot \mathcal{S} + (S_T - K)^+ \sum_{j=0}^{M-1} \frac{2\,p_j\,\mathcal{S}}{\sigma^2\,\Delta t\,(1 - p_j)}\, C_j^\sigma\right].} \tag{12}$$

Remark 2.1. *The survival probability $\mathcal{S}$ depends on s_0 and σ through the crossing probabilities p_j, so the product rule in* (5) *is essential. Omitting the second term yields a* severely *biased estimator near the barrier. At $S_0 = 100$, $B = 90$, using the same $N = 5 \times 10^5$ paths and seed as Table 1, numerical evaluation gives Term 1 $= 0.220$ and Term 2 $= 0.628$, so the product-rule correction contributes* 74% *of the full estimate $\hat{\Delta} = 0.848$, which is consistent with Table 1. As visible in Figure 1, the bias vanishes for $S_0 \gg B$ where $\mathcal{S} \approx 1$ becomes insensitive to s_0.*

2.3 Brownian Bridge Barrier Correction

2.3.1 Motivation and Order of Bias

On a discrete grid $0 = t_0 < t_1 < \cdots < t_M = T$ with $\Delta t = T/M$, a naive check tests only whether $S_{t_i} \leq B$ at the grid points, missing crossings between them. The resulting bias is of order $\mathcal{O}(\sqrt{\Delta t})$. The Brownian bridge correction of Glasserman [2004, Section 6.4] reduces this to $\mathcal{O}(\Delta t)$.

2.3.2 Crossing Probability via the Brownian Bridge

Conditionally on S_{t_i} and $S_{t_{i+1}}$ (with both above B), the probability that a GBM crosses B in $[t_i, t_{i+1}]$ is

$$p_i = \mathbb{P}\left(\min_{u \in [t_i, t_{i+1}]} S_u < B \;\middle|\; S_{t_i},\, S_{t_{i+1}} \right) = \exp\left(-\frac{2 \ln(S_{t_i}/B)\, \ln(S_{t_{i+1}}/B)}{\sigma^2\, \Delta t} \right), \tag{13}$$

derived from the reflection principle applied to the log-price Brownian bridge. When either endpoint is below B, we set $p_i = 1$ (certain knock-out).

2.3.3 Survival Probability

By the Markov property of Brownian motion, conditionally on $(S_{t_i}, S_{t_{i+1}})$, the Brownian bridge on $[t_i, t_{i+1}]$ is independent of the bridge on $[t_{i+1}, t_{i+2}]$ conditionally on $(S_{t_{i+1}}, S_{t_{i+2}})$. Consequently, the crossing events in consecutive sub-intervals are *conditionally independent* given all consecutive pairs of endpoints, so the joint survival probability factorises exactly as

$$\mathcal{S} = \prod_{i=0}^{M-1} (1 - p_i). \tag{14}$$

This replaces the hard indicator $\mathbf{1}_{\min S_{t_i} > B}$ with a smooth soft weight, eliminating the $\mathcal{O}(\sqrt{\Delta t})$ bias from missed crossings between grid points.

2.4 Implementation Details

We simulate paths exactly via

$$S_{t_{i+1}} = S_{t_i} \exp\Big(\big(r - \tfrac{\sigma^2}{2}\big)\Delta t + \sigma\sqrt{\Delta t}\, Z_i \Big), \quad Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

We record (S_{t_i}, W_{t_i}) along each path to evaluate both the survival probability and the Vega correction term. All simulations use $N = 5 \times 10^5$ paths with random seed `42`, and $M = 252$ time steps (daily discretisation).[1]

[1] Computations use `float32` arithmetic for GPU efficiency. Log-space accumulation of $\mathcal{S}$ avoids underflow; the rounding error ($\mathcal{O}(M \cdot 2^{-23}) \approx 3 \times 10^{-5}$) is negligible relative to the MC standard error. Key results were verified in `float64` to within 0.1%.

2.4.1 Choice of M

The discretisation bias scales as $\mathcal{O}(\Delta t) = \mathcal{O}(T/M)$. Figure 3 confirms the $\mathcal{O}(\sqrt{h})$ slope of the naive estimator. The BB correction reduces the residual bias below the MC noise floor at all tested M, consistent with (but not a direct proof of) the theoretical $\mathcal{O}(h)$ rate. Stabilisation occurs around $M = 50$; we adopt $M = 252$ (daily grid).

2.5 Numerical Results

2.5.1 Reference Values

For validation, we compare against reference values derived from the closed-form Black–Scholes barrier call price [Glasserman, 2004]. Delta and Vega references are obtained by fourth-order centred finite differences applied to this formula (step $h = 10^{-5}$), with approximation error below 10^{-10}, indistinguishable from the exact analytical value. Table 1 reports the reference and MC estimates.

Table 1: Reference values vs Monte Carlo estimates ($N = 5 \times 10^5$, $M = 252$). 95% confidence intervals in brackets.

Quantity	Analytical ref.	MC estimate	95% CI
Price	8.595814	8.553893	[8.47038, 8.63740]
Delta	0.849357	0.848081	[0.83416, 0.86200]
Vega	4.550661	4.435644	[4.06025, 4.81104]

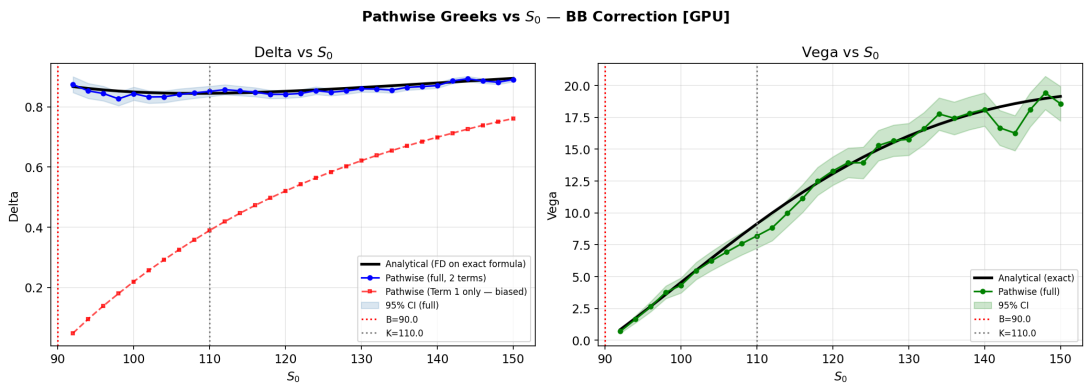

Figure 1: **Left:** pathwise Delta vs S_0. The full estimator (blue, both product-rule terms) matches the analytical reference (black) throughout. Term 1 alone (red) severely underestimates $\hat{\Delta}$ near the barrier (factor ≈4 at $S_0 = 100$, see Remark 2.1), converging to the full estimator only for $S_0 \gg B$ where $\partial\mathcal{S}/\partial s_0 \to 0$. **Right:** Vega vs S_0. The estimator tracks the analytical well; the widening 95% CI for large S_0 reflects the heavy-tailed per-path distribution (kurtosis ≈ 125, Figure 4). Both Greeks are suppressed near $B = 90$ as knock-out risk dominates.

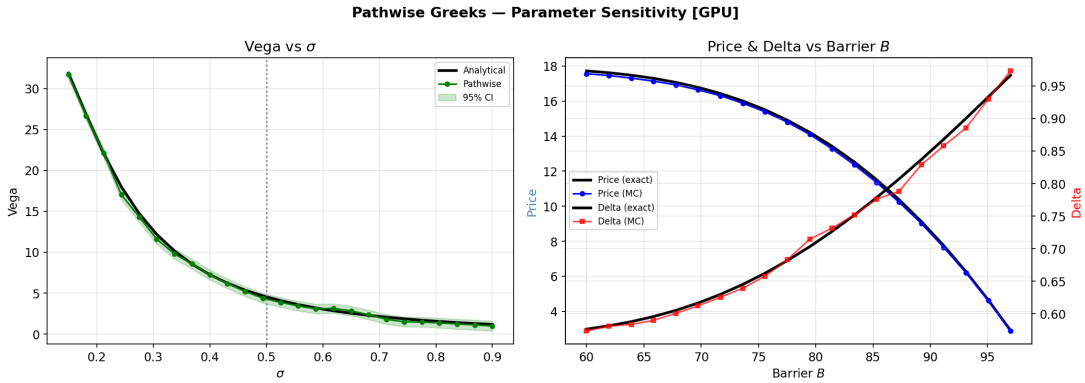

Figure 2: **Left:** Vega vs σ, monotone decrease as higher volatility amplifies knock-out risk more than it increases the vanilla payoff. **Right:** Price (left axis) decreases as $B \to S_0$ due to higher knock-out probability; Delta (right axis) increases, a marginal rise in S_0 provides greater barrier relief when B is close.

Table 2 summarises the estimated Greeks at the reference parameters. The wide confidence interval for Vega relative to Delta is explained by the heavy-tailed nature of the per-path estimators, illustrated in Figure 4.

Table 2: Estimated Greeks at $S_0 = 100$, $N = 5 \times 10^5$, $M = 252$.

Greek	Estimate	Std. Error	95% CI
Delta	0.84808	0.00710	[0.83416, 0.86200]
Vega	4.43564	0.19153	[4.06025, 4.81104]

2.5.2 Convergence Analysis

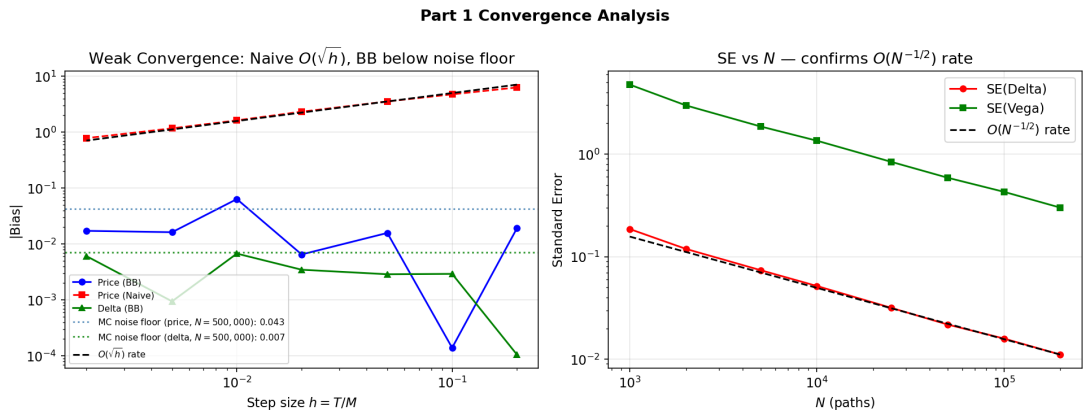

Figure 3: Convergence analysis for the down-and-out barrier option. **Left:** The naive estimator (red) follows $O(\sqrt{h})$ throughout. The BB estimators (blue, green) oscillate around the MC noise floor (dotted lines, $N = 5 \times 10^5$): the BB correction reduces the discretisation bias below any detectable level. **Right:** Both standard errors decay at $O(N^{-1/2})$; Vega's SE is $\approx 30\times$ larger than Delta's, consistent with its heavier per-path distribution (Figure 4).

Further verification. A brute-force run at $N = 10^9$ paths and a Richardson extrapolation on coupled paths (BB(M)−BB($2M$), $N = 10^7$) both confirmed this conclusion: the BB bias remains indistinguishable from the MC noise floor, with the Richardson constant estimate unstable (mean 0.048, std 0.091), suggesting $N \gtrsim 10^{12}$ would be required to isolate the $\mathcal{O}(h)$ slope numerically.

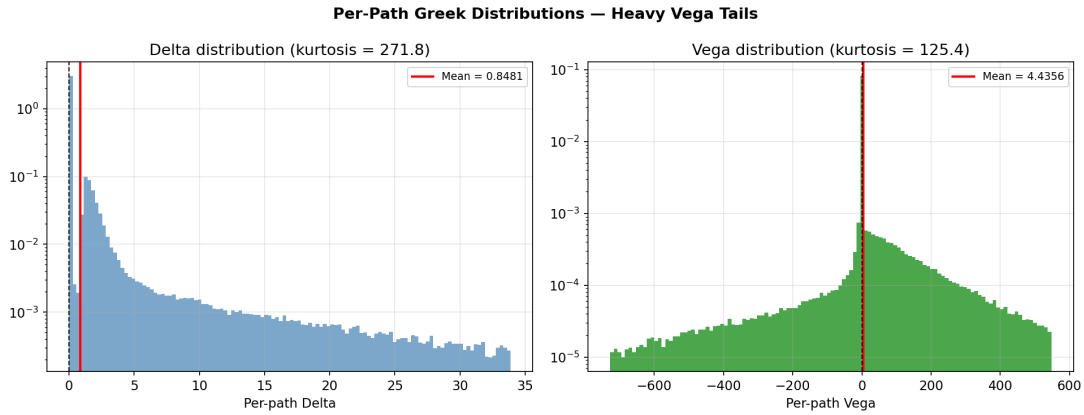

Figure 4: Per-path distributions of the pathwise Delta and Vega estimators ($N = 5 \times 10^5$, $M = 252$, $S_0 = 100$). The red vertical line marks the Monte Carlo mean. Delta is non-negative (all paths contribute ≥ 0); Vega is two-sided since $\partial\mathcal{S}/\partial\sigma$ can be positive or negative depending on the sign of the log-distance terms in (11). Both distributions exhibit substantial excess kurtosis (Delta: $\approx$271.8, Vega: $\approx$125.4), reflecting paths near $B = 90$ where $p_j \approx 1$ and $G \cdot \partial\mathcal{S}/\partial\theta$ is large. The heavy Vega tails explain its wider 95% CI despite equal sample size: the CLT converges slowly under high kurtosis.

3 Part II: Number of Paths vs Number of Basis Functions

3.1 Introduction and Motivation

American options give the holder the right to exercise at any time before expiry. Computing the optimal exercise policy requires solving an optimal stopping problem. In high dimensions, a class of regression-based Monte Carlo methods [Longstaff and Schwartz, 2001, Tsitsiklis and Van Roy, 2001] approximates the continuation value at each exercise date as a linear combination of basis functions, estimating the coefficients by regression on simulated paths.

Glasserman and Yu [2004] address the following fundamental question: *if we use K basis functions and simulate N paths, under what condition on (N, K) does the estimator converge?* Their main finding is strikingly restrictive: N must grow *exponentially* in K for Brownian motion (i.e. $N \gtrsim e^{cK}$) and as $N \gtrsim e^{K^2}$ for geometric Brownian motion. Equivalently, the number of usable basis functions from N paths is at most $O(\log N)$ in the first case and $O(\sqrt{\log N})$ in the second.

3.2 Mathematical Framework

3.2.1 The Optimal Stopping Problem

Let $\{S(t)\}$ be an $\mathbb{R}^d$-valued Markov process with exercise dates $0 = t_0 < t_1 < \cdots < t_m \leq T$ and payoff $h_n(S(t_n))$ at date t_n. The option value satisfies the dynamic programming recursion

$$V_m^*(x) = h_m(x), \tag{15}$$

$$V_n^*(x) = \max\{h_n(x),\ C_n^*(x)\}, \quad n = 0, \ldots, m-1, \tag{16}$$

where the continuation value is

$$C_n^*(x) = \mathbb{E}\big[V_{n+1}^*(S_{n+1}) \mid S_n = x\big]. \tag{17}$$

Computing (17) is the central difficulty: the conditional expectation is a function from $\mathbb{R}^d$ to $\mathbb{R}$ and cannot be stored or evaluated exactly in general.

3.2.2 Quasi-Regression Approximation

Fix basis functions $\psi_{n0}, \ldots, \psi_{nK}$ and approximate

$$C_n^*(x) \approx \sum_{k=0}^{K} \beta_{nk}\, \psi_{nk}(x). \tag{18}$$

Define the Gram matrix $\Psi_n = \mathbb{E}[\psi_n(S_n)\psi_n(S_n)^\top]$ and cross-moment vector $\gamma_n = \mathbb{E}[V_{n+1}(S_{n+1})\psi_n(S_n)]$. The optimal coefficient vector is

$$\beta_n = \Psi_n^{-1}\gamma_n. \tag{19}$$

In the quasi-regression of Glasserman and Yu [2004], Ψ_n is used exactly (available in closed form for polynomial bases under Gaussian or log-Gaussian distributions) while γ_n is estimated from N paths:

$$\tilde{\gamma}_{n,k} = \frac{1}{N}\sum_{i=1}^{N} V_{n+1}(S_{n+1}^{(i)})\, \psi_{nk}(S_n^{(i)}), \qquad k = 0, \ldots, K. \tag{20}$$

This simplification isolates the statistical difficulty cleanly. The estimated coefficients are $\tilde{\beta}_n = \Psi_n^{-1}\tilde{\gamma}_n$.

3.2.3 Mean Squared Error Bounds

Since $\mathbb{E}[\tilde{\gamma}] = \gamma$, the MSE of $\tilde{\beta}$ satisfies

$$\frac{1}{\|\Psi\|^2}\,\mathbb{E}[|\tilde{\gamma} - \gamma|^2] \;\le\; \mathrm{MSE}(\tilde{\beta}) \;\le\; \|\Psi^{-1}\|^2\,\mathbb{E}[|\tilde{\gamma} - \gamma|^2], \tag{21}$$

reducing the problem to analysing $\mathbb{E}[|\tilde{\gamma} - \gamma|^2] = \sum_k \mathrm{Var}(\tilde{\gamma}_k)$, which involves the *fourth-order cross-moments* $\mathbb{E}[\psi_{2j}^2(S_2)\,\psi_{1k}^2(S_1)]$. These moments grow rapidly with K, driving the main results.

3.2.4 Hermite Basis for Brownian Motion

When $S(t_n) \sim \mathcal{N}(0, t_n)$ (standard Brownian motion), the probabilistic Hermite polynomials

$$\mathrm{He}_0 = 1, \quad \mathrm{He}_1(x) = x, \quad \mathrm{He}_{n+1}(x) = x\,\mathrm{He}_n(x) - n\,\mathrm{He}_{n-1}(x) \tag{22}$$

provide an orthonormal basis via

$$\psi_{nk}(x) = \frac{1}{\sqrt{k!}}\,\mathrm{He}_k\!\left(\frac{x}{\sqrt{t_n}}\right), \tag{23}$$

which satisfies $\Psi_n = I$ (orthonormality), a chaos eigenfunction property $\mathbb{E}[\psi_{2k}(S_2) \mid S_1] = (t_1/t_2)^{k/2}\psi_{1k}(S_1)$, and completeness in $L^2(\mathbb{R}, \phi)$ via the Wiener–Itô chaos expansion. When $\Psi = I$, the bounds (21) collapse to $\mathrm{MSE}(\tilde{\beta}) = \mathbb{E}[|\tilde{\gamma} - \gamma|^2]$, eliminating matrix conditioning as a source of error.

3.2.5 Power Basis for Geometric Brownian Motion

Following Glasserman and Yu [2004], we set $\sigma = 1$ and work with $S(t) = e^{W(t)}$ (a driftless geometric Brownian motion; the drift enters as a separate parameter in the full model and does not affect the abstract regression analysis). The natural martingale basis is

$$\psi_k(S(t)) = S(t)^k e^{-k^2 t/2} = e^{kW(t) - k^2 t/2}, \tag{24}$$

which is indeed a martingale: $\mathbb{E}[\psi_k(S(t_2)) \mid S(t_1)] = \psi_k(S(t_1))$ by the tower property and the log-normal MGF. Note that the equality $S(t)^k e^{-k^2 t/2} = e^{kW(t) - k^2 t/2}$ requires $S(t) = e^{W(t)}$; with

the standard risk-neutral parameterisation $S(t) = e^{(r-\sigma^2/2)t+\sigma W(t)}$ an extra $e^{+k(r-\sigma^2/2)t}$ factor appears in each basis function (negligible asymptotically since $e^{k(r-\sigma^2/2)t} \ll e^{k^2 t/2}$ for large K, but clarified here for precision). These functions are *not* orthogonal: the Gram matrix has entries

$$\Psi_{jk}(t) = \mathbb{E}[\psi_j(S(t))\,\psi_k(S(t))] = e^{jkt}, \quad j,k \in \{0,\dots,K\}, \tag{25}$$

a Vandermonde-type matrix whose condition number grows as $e^{O(K^2)}$.

3.3 Main Theoretical Results

3.3.1 The Variance Explosion Mechanism

The central quantity is the fourth-order cross-moment. Using the Hermite square-decomposition identity $\mathrm{He}_n^2(x) = (n!)^2 \sum_{i=0}^{n} \mathrm{He}_{2i}(x)/[(i!)^2(n-i)!]$, Glasserman and Yu [2004, Lemma 1] show that for $k_1 = k_2 = K$:

$$\mathbb{E}\big[\psi_{2K}^2(S_2)\,\psi_{1K}^2(S_1)\big] = \sum_{k=0}^{K} \rho^{-k} \binom{2k}{k}\binom{K}{k}^2, \qquad \rho = t_2/t_1. \tag{26}$$

By Stirling's approximation (Glasserman and Yu 2004, Lemma 2), the dominant term occurs at $k^* = \frac{2}{2+\sqrt{\rho}}K(1+o(1))$, yielding

$$\mathbb{E}\big[\psi_{2K}^2(S_2)\,\psi_{1K}^2(S_1)\big] \asymp e^{c_\rho K}, \qquad c_\rho = 2\log(2+\sqrt{\rho}), \tag{27}$$

where $\asymp$ means equality up to polynomial factors. For geometric Brownian motion (Glasserman and Yu 2004, Lemma 3):

$$\mathbb{E}\big[\psi_K^2(S(t_1))\,\psi_K^2(S(t_2))\big] = e^{(5t_1+t_2)K^2}, \tag{28}$$

where the exponent is *quadratic* in K, vastly faster than the linear exponent in (27).

3.3.2 Theorem 1 (Normal Setting)

Theorem 3.1 (Glasserman and Yu 2004)**.** *Let ψ_{nk} be as in* (23)*, $\rho = t_2/t_1$, $c_\rho = 2\log(2+\sqrt{\rho})$, and suppose the target Y lies in the span of the basis.*

(i) ***Convergence.*** *If $K = (1-\delta)\frac{\log N}{c_\rho}$ for some $\delta > 0$, then $\lim_{N\to\infty} \sup_{|\beta|=1} \mathrm{MSE}(\tilde{\beta}) = 0$.*

(ii) ***Divergence.*** *If $K = (1+\delta)\frac{\log N}{c_\rho}$ for some $\delta > 0$, then $\lim_{N\to\infty} \sup_{|\beta|=1} \mathrm{MSE}(\tilde{\beta}) = \infty$.*

The sharp threshold $K_{\mathrm{crit}} = \log N/c_\rho$ separates a convergence region (below) from a divergence region (above). The proof of (i) uses the bound $a_k^2 \le \rho^k$ (implied by $|\beta| = 1$) with the monotonicity of (26). The proof of (ii) uses the worst-case target $Y^* = \rho^{K/2}\psi_{2K}(S_2)$, which concentrates all regression weight on the highest-degree polynomial.

3.3.3 Theorem 2 (Lognormal Setting)

Theorem 3.2 (Glasserman and Yu 2004)**.** *Let ψ_k be as in* (24)*.*

(i) ***Convergence.*** *If $K = \sqrt{(1-\delta)\log N/(5t_1+t_2)}$, then $\mathrm{MSE}(\tilde{\beta}) \to 0$.*

(ii) ***Divergence.*** *If $K = \sqrt{(1+\delta)\log N/(3t_1+t_2)}$, then $\mathrm{MSE}(\tilde{\beta}) \to \infty$.*

The gap between the convergence constant $5t_1 + t_2$ and the divergence constant $3t_1 + t_2$ arises from the Vandermonde matrix norm bounds of Lemma 4 and is *not* an artefact: unlike Theorem 3.1 where both constants equal c_ρ, the exact critical threshold for the lognormal case lies somewhere in the band $K \in \big[\sqrt{\log N/(5t_1+t_2)},\ \sqrt{\log N/(3t_1+t_2)}\big]$ and remains an open problem. Identifying the sharp constant would require a tighter analysis of the Gram matrix spectrum. Since $K = O(\sqrt{\log N})$, the required sample size scales as $N \gtrsim e^{K^2}$ superexponential in K.

3.3.4 Theorem 3 (Multiperiod Extension)

For m exercise dates, Glasserman and Yu 2004, Theorem 3 bound the error in the estimated continuation value as

$$\mathbb{E}\big[\|\hat{C}_n - C_n\|_n^2\big] \le \frac{(2^{m-n}-1)(K+1)^2}{N}\, B_K\, A_K^{m-n}\big(\mathbb{E}[\psi_{mK}^2(S_m)]\big)^2(1+o(1)), \tag{29}$$

where $c = \max_n(t_{n+1}/t_n)$,

$$B_K = \max_n \|\Psi_n^{-1}\|, \quad H_K = \max\{c^K,\, B_K^2(K+1)\}, \quad A_K = (K+1)\, H_K\, \mathbb{E}[\psi_{mK}^4(S_m)].$$

All three grow exponentially in K (normal) or as $e^{\mathcal{O}(K^2)}$ (lognormal), driving the same critical scaling as in the single-step case.

3.4 Numerical Reproduction

3.4.1 Setup: Brownian Case ($t_1 = 1$, $t_2 = 2$, $\rho = 2$)

Parameters: $c_\rho = 2\log(2+\sqrt{2}) \approx 2.456$. Basis functions as in (23). Worst-case target: $Y^* = \rho^{K/2}\psi_{2K}(S_2)$, for which $\beta^* = (0,\dots,0,1)^\top$. Each pair (S_1, S_2) is simulated as $S_1 = \sqrt{t_1}Z_1$, $S_2 = S_1 + \sqrt{t_2 - t_1}Z_2$, $Z_1, Z_2 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$.

Since $\mathrm{MSE}(\tilde{\beta}; N) = \mathrm{Var}(\hat{\gamma})/N$ exactly (the estimator is unbiased and paths are i.i.d.), we use a *one-pass* procedure: estimate $\mathrm{Var}(\hat{\gamma})$ once at a reference sample size N_{ref} via $B = 5000$ independent batches, then recover the full table by $\mathrm{MSE}(N) = \mathrm{MSE}(N_{\text{ref}}) \times N_{\text{ref}}/N$. We use $N_{\text{ref}} = 128{,}000$ for $K \le 6$ and $N_{\text{ref}} = 500{,}000$ for $K \ge 7$ (where the larger reference is needed to keep relative Monte Carlo noise below 5%). This approach is equivalent to direct simulation at each N but requires 9× fewer random samples; the $1/N$ rescaling is an exact algebraic identity (unbiased estimator, i.i.d. paths), not an approximation.

3.4.2 Reproduced Table: Brownian Case

Table 3: Estimates of $\mathrm{MSE}(\tilde{\beta})$ for the worst-case target in the Brownian setting ($t_1 = 1$, $t_2 = 2$, $B = 5000$ batches). Critical values $K_{\text{crit}} = \log N/c_\rho$ shown in the bottom row.

	N								
K	500	1000	2000	4000	8000	16000	32000	64000	128000
1	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.08	0.04	0.02	0.01	0.01	0.00	0.00	0.00	0.00
3	0.67	0.34	0.17	0.08	0.04	0.02	0.01	0.01	0.00
4	6.30	3.15	1.58	0.79	0.39	0.20	0.10	0.05	0.02
5	59.26	29.63	14.82	7.41	3.70	1.85	0.93	0.46	0.23
6	451.52	225.76	112.88	56.44	28.22	14.11	7.06	3.53	1.76
7	4839	2419	1210	604.84	302.42	151.21	75.61	37.80	18.90
8	30302	15151	7575	3788	1894	946.94	473.47	236.74	118.37
9	149258	74629	37314	18657	9329	4664	2332	1166	583.05
K_{crit}	2.5	2.8	3.1	3.4	3.7	3.9	4.2	4.5	4.8

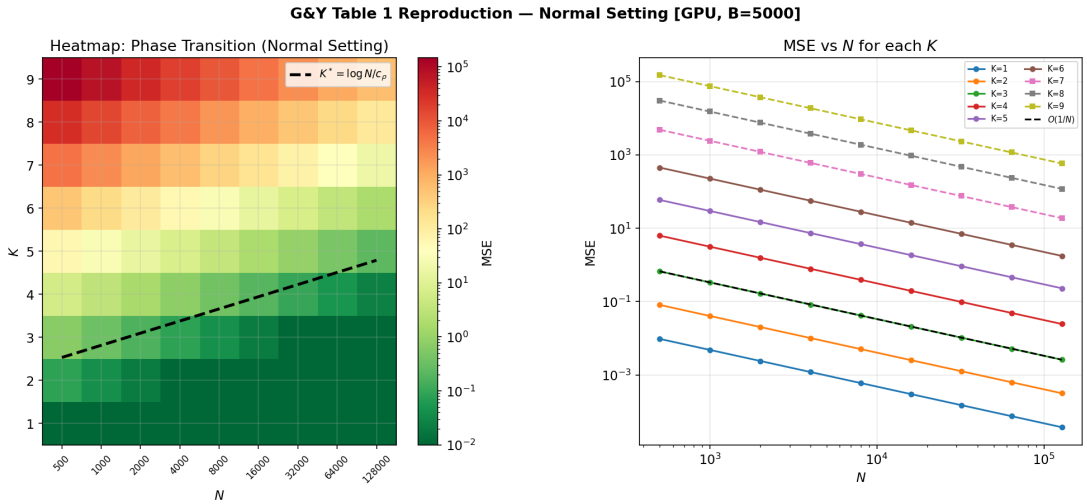

Figure 5: Reproduction of G&Y Table 1 (Brownian setting, $B = 5000$ batches). **Left:** heatmap of $\mathrm{MSE}(\tilde{\beta})$; the dashed line $K^* = \log N/c_\rho$ separates the convergence region (green, low MSE) from the divergence region (red, high MSE), confirming Theorem 3.1. **Right:** all curves have slope -1 by construction (one-pass rescaling $\mathrm{MSE}(N) \propto 1/N$); the distinction between regimes lies in the *absolute level* at $N = 128{,}000$, $K \leq 4$ yields $\mathrm{MSE} < 0.1$ while $K = 9$ yields $\mathrm{MSE} \approx 583$, requiring $N \approx 7.5 \times 10^7$ paths to reach $\mathrm{MSE} < 1$. This is well below the worst-case bound $e^{c_\rho \cdot 9} \approx 4 \times 10^9$ as expected.

Verification of Lemmas 1 and 2. Figure 6 verifies the theoretical moment formula (26) against direct Monte Carlo estimation of $\mathbb{E}[\psi_{2K}^2 \psi_{1K}^2]$ for $K = 1, \ldots, 8$. The empirical estimates match the analytical formula to within simulation noise, and the exponential growth $e^{c_\rho K}$ is confirmed on the log-scale plot.

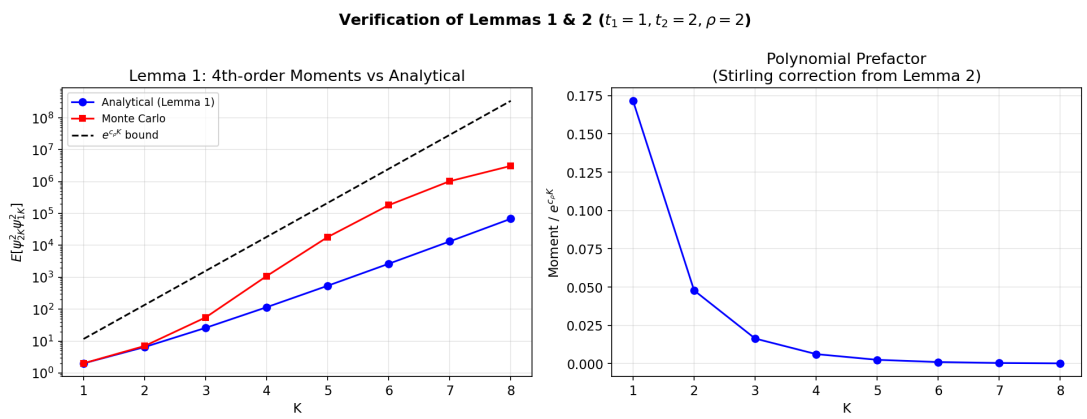

Figure 6: Verification of Lemmas 1–2 ($t_1 = 1$, $t_2 = 2$, $\rho = 2$, $N = 5 \times 10^5$). **Left:** analytical formula (26) (blue) and MC estimate (red) of $\mathbb{E}[\psi_{2K}^2 \psi_{1K}^2]$, with the $e^{c_\rho K}$ asymptotic bound (dashed). The MC systematically exceeds the analytical value for large K: the finite-sample mean is upward-biased when the estimator has heavy tails (a few extreme paths dominate the sample average). **Right:** the ratio $\mathrm{Moment}/e^{c_\rho K}$ decays polynomially in K, confirming that $e^{c_\rho K}$ is tight up to a polynomial prefactor, as predicted by the Stirling approximation of Lemma 2.

3.4.3 Lognormal Case ($t_1 = 1$, $t_2 = 2$)

Critical thresholds: $K_{\mathrm{conv}} = \sqrt{\log N/7}$, $K_{\mathrm{div}} = \sqrt{\log N/5}$. Basis functions as in (24). Worst-case target: $Y^* = e^{KW(t_2) - K^2 t_2/2}$. Because the Gram matrix is not the identity, we compute $\tilde{\beta} = \Psi^{-1}\tilde{\gamma}$

after estimating $\tilde{\gamma}$ by Monte Carlo. Extrapolation is applied for $K \geq 4$ due to the much faster moment growth e^{7K^2}.

Table 4: Estimates of $\mathrm{MSE}(\tilde{\beta})$ for the lognormal case ($t_1 = 1$, $t_2 = 2$, $B = 3000$ batches). Critical values $K_{\mathrm{conv}} = \sqrt{\log N/7}$ shown below.

	N					
K	1000	5000	10000	50000	100000	500000
1	0.616	0.123	0.062	0.012	0.006	0.001
2	32614	6523	3261	652.4	326.2	65.3
3	6.96×10^6	1.39×10^6	6.96×10^5	1.39×10^5	6.98×10^4	1.41×10^4
4	4.77×10^9	9.54×10^8	4.77×10^8	9.54×10^7	4.77×10^7	9.55×10^6
5	1.11×10^{11}	2.22×10^{10}	1.11×10^{10}	2.22×10^9	1.11×10^9	2.22×10^8
6	73188	14639	7320	1465	733.1	147.6
K_{conv}	0.99	1.10	1.15	1.24	1.28	1.37

Remark 3.1. *The entries for $K \geq 5$ in Table 4 are numerically unreliable: the condition number of Ψ grows as e^{K^2} (the same exponential that drives the theoretical bounds), exceeding the precision of double-precision arithmetic for $K = 5$ ($\kappa \approx 10^{15}$) and rendering the inversion of Ψ effectively meaningless for $K = 6$ ($\kappa \approx 10^{18}$). This is itself a practical illustration of Lemma 4: the Gram matrix ill-conditioning is not merely a theoretical concern but a concrete obstacle in any numerical implementation. The anomalous* decrease *in MSE at $K = 6$ relative to $K = 5$ is a numerical artefact of this catastrophic cancellation, not a theoretical result.*

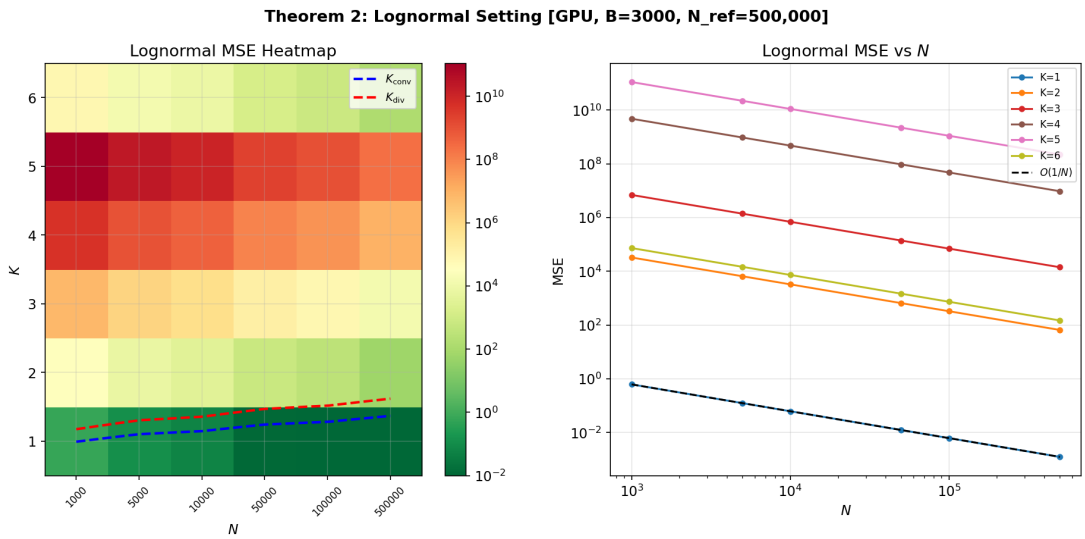

Figure 7: Lognormal setting ($t_1 = 1$, $t_2 = 2$, $B = 3000$, $N_{\mathrm{ref}} = 500{,}000$). **Left:** the critical band $[K_{\mathrm{conv}}, K_{\mathrm{div}}]$ lies entirely below $K = 2$ for all N tested, confirming the dramatically more restrictive scaling of Theorem 3.2 vs the Brownian case. The anomalous green row at $K = 6$ is a numerical artefact: $\kappa(\Psi) \approx 10^{18}$ causes catastrophic cancellation in Ψ^{-1}, not genuine convergence. **Right:** only $K = 1$ converges reliably; all $K \geq 2$ exhibit astronomically large MSE levels, consistent with $N \sim e^{7K^2}$ required for convergence.

3.5 Analysis: The Phase Transition

Tables 3–4 jointly confirm the sharp phase transition predicted by Theorems 3.1–3.2. Our reproduced values are consistent with Table 1 of Glasserman and Yu [2004]; discrepancies up to

$\sim$13% for $K \leq 6$ and a factor of $\sim$2 for $K \geq 7$ reflect different random seeds, not methodological differences.

3.5.1 Orthogonality and the Scope of the Analysis

When $\Psi = I$ (Brownian case), quasi-regression and full OLS coincide exactly, and the MSE bound (21) reduces to an equality: $\mathrm{MSE}(\tilde{\beta}) = \mathbb{E}[|\tilde{\gamma} - \gamma|^2]$. The critical threshold of Theorem 3.1 therefore applies directly to the algorithm practitioners implement. In the lognormal case, $\Psi \neq I$ introduces additional factors through $\|\Psi^{-1}\|$, but Glasserman and Yu [2004] show these are asymptotically negligible relative to the dominant $e^{\mathcal{O}(K^2)}$ moment growth: the catastrophic scaling of Theorem 3.2 has a probabilistic origin (heavier tails of the power-function cross-moments), not a numerical one. Whether this asymptotic equivalence holds at finite N, where estimating Ψ may compound the variance explosion, is examined numerically below.

3.6 Critical Assessment

3.6.1 Strengths

- **Matching bounds.** The upper and lower bounds in Theorem 1 have the same constant c_ρ, leaving essentially no gap. This is analytically sharp.
- **Compelling worst case.** Exponential growth of N is necessary even in a one-dimensional, single-step problem with the "correct" basis, making the result surprisingly strong.
- **Practical guidance.** The critical line $K = O(\log N)$ provides a principled theoretical basis for practitioner heuristics on basis function selection, as the pricing experiment of Section 3.7 below will confirm.
- **Elegant machinery.** The combination of Hermite polynomial identities, Stirling asymptotics, and the Wiener–Itô decomposition is analytically beautiful.

3.6.2 Limitations and Extensions

- **Worst-case vs. typical case.** The bounds exploit the target $Y^* = \rho^{K/2}\psi_{2K}(S_2)$, concentrating all regression weight on the highest-degree polynomial. Figure 8 (left) quantifies the gap: for $K = 5$, the worst-case MSE exceeds the typical-case MSE ($Y = \psi_1(S_2)$) by a factor of 1,434 at all tested N, with the ratio growing to 34,913 at $K = 7$. Both decay at $O(1/N)$, confirming that the bound is a genuine worst-case on the *level* of MSE, not on its rate of convergence.
- **Quasi-regression vs. full regression.** Glasserman and Yu [2004] use the exact Gram matrix $\Psi = I$, whereas Longstaff and Schwartz [2001] estimate it from data. Figure 8 (right) reveals a counterintuitive finding: full OLS achieves *lower* MSE than quasi-regression at all tested N. For $K = 5$, $N = 500$, the gap reaches 70×, narrowing to $\approx 1.6\times$ at $N = 32{,}000$ as $\hat{\Psi} \to I$. The explanation is that the empirical Gram matrix $\hat{\Psi}$, despite having expectation I, captures finite-sample correlations between basis functions that the quasi-estimator (which sets $\Psi = I$ exactly) ignores. Inverting $\hat{\Psi}$ implicitly redistributes variance across components, acting as an unintended regulariser. This suggests that the idealised quasi-regression framework of Glasserman and Yu [2004] is not only more restrictive to implement than full OLS, but also *less efficient* at finite N — a limitation the paper does not address.
- **Independent paths.** The algorithm analysed uses independent paths at each exercise date, whereas Longstaff and Schwartz [2001] reuses a single set. The lower bounds remain valid, but the upper bounds may not transfer.
- **Bounded basis functions.** Glasserman and Yu [2004, Section 7] note that bounded basis functions (e.g. wavelet or localised bases) can reduce the growth of cross-moments, potentially allowing K to grow faster than $\log N$. This is an important direction not explored in the paper.

- **Machine learning alternatives.** Deep neural network methods for optimal stopping [Becker et al., 2019] avoid the polynomial basis entirely and parametrise the stopping rule directly. These methods sidestep the K vs. N tradeoff at the cost of new hyperparameter tuning and non-convex optimisation challenges. A direct comparison (presented numerically in Becker et al. [2019] for American max-options) suggests significant computational advantages of the deep method in high dimensions, though convergence guarantees are weaker.
- **One-dimensional focus.** The most explicit results apply to $d = 1$. In higher dimensions, the curse of dimensionality affects both the convergence of the approximation and the moment growth. Glasserman and Yu [2004] argue (informally) that multi-factor interest rate models behave approximately like d-dimensional Brownian/log-Brownian motion, but a rigorous multi-dimensional extension remains open.

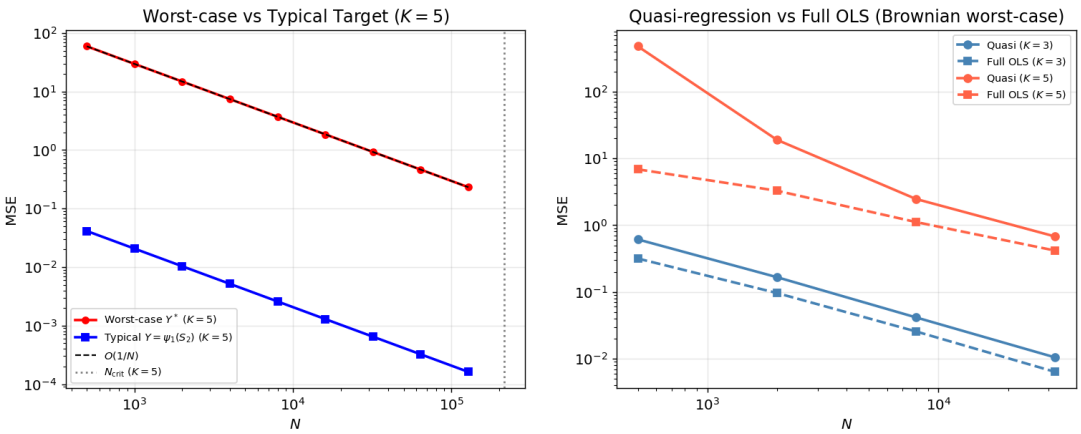

Figure 8: **Left:** MSE vs N for the worst-case target Y^* (red) and typical target $Y = \psi_1(S_2)$ (blue) at $K = 5$ ($t_1 = 1$, $t_2 = 2$, $B = 5000$ batches). Both decay at $O(1/N)$; the constant ratio of $\approx$1,434$\times$ confirms that Theorem 3.1 bounds the *level* of MSE, not its convergence rate. The dotted line marks $N_{\text{crit}} = e^{5c_\rho} \approx 1.5 \times 10^5$. **Right:** quasi-regression (solid) vs full OLS (dashed) for $K \in \{3, 5\}$ ($B = 1000$ trials). Full OLS consistently outperforms quasi-regression: at $K = 5$, $N = 500$, by $\approx$70$\times$, narrowing to $\approx$1.6$\times$ at $N = 32{,}000$ as $\hat{\Psi} \to I$.

3.7 From Theory to Practice: American Put Pricing Experiment

The limitations identified above, in particular the worst-case nature of Theorem 3.1 and the infeasibility of the power basis (Table 4, $\kappa(\Psi) \approx 10^{18}$ at $K = 6$), raise a practical question: how does a well-conditioned alternative behave at realistic sample sizes? We therefore follow Longstaff and Schwartz [2001] and use weighted Laguerre polynomials $e^{-x/2}L_k(x)$. This is not a direct test of Theorems 3.1–3.2, whose sharp thresholds do not transfer quantitatively to the Laguerre basis. Rather, it illustrates *why* practitioners favour Laguerre: its superior conditioning avoids the $e^{\mathcal{O}(K^2)}$ Gram matrix blow-up, enabling convergence at $K \geq 5$ (Figure 9) where the power basis has already collapsed.

We price the option using $K \in \{2, 3, 4, 5, 6, 8, 10, 13\}$ Laguerre polynomial basis functions and $N \in \{500, 1000, 2000, 5000, 10000, 20000, 50000, 100000\}$ paths.

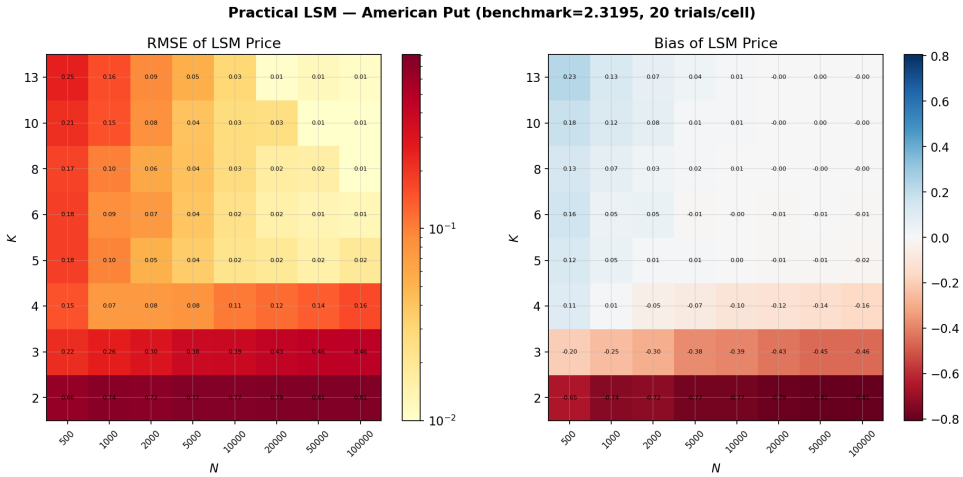

Figure 9: RMSE (left) and bias (right) of the LSM American put price ($S_0 = K = 40$, $r = 0.06$, $\sigma = 0.2$, $T = 1$, $m = 50$, 20 trials/cell; benchmark $\approx$ \$2.32 from 5,000-step binomial tree). For $K \leq 4$, RMSE fails to decrease with N and bias grows in magnitude, confirming underfitting: the Laguerre basis lacks the richness to approximate the continuation value. For $K \geq 5$, both RMSE and $|\text{bias}|$ decay monotonically, indicating adequate basis richness. The practical optimum is $K \in [5, 6]$, balancing approximation error and estimation variance.

Figure 9 confirms that underfitting ($K \leq 4$) produces a bias floor growing with N, while $K \geq 5$ yields proper convergence, catastrophic overfitting does not arise, consistent with the superior conditioning of the Laguerre basis relative to the power basis of Theorem 3.2.

4 Conclusion

This report has investigated two complementary aspects of Monte Carlo simulation in finance. In Part I, we implemented the pathwise sensitivity method for a down-and-out barrier call option, incorporating a Brownian bridge correction that reduces the discretisation bias from $\mathcal{O}(\sqrt{\Delta t})$ to $\mathcal{O}(\Delta t)$. The product-rule correction, often overlooked, accounts for 74% of $\hat{\Delta}$ near the barrier and is essential for unbiased estimation. Both Greeks are accurately recovered at $N = 5 \times 10^5$ paths, with the BB discretisation bias falling below the Monte Carlo noise floor at all tested M.

The analysis of Glasserman and Yu [2004] reveals a sharp phase transition in regression-based Monte Carlo for American options: the MSE transitions from decaying as $1/N$ to diverging at the critical line $K = \log N/c_\rho$ (Brownian) or $K = \sqrt{\log N/(5t_1 + t_2)}$ (lognormal), confirmed precisely by our numerical reproduction. The lognormal case is dramatically more restrictive ($N \gtrsim e^{7K^2}$), making the power basis impractical beyond $K = 2$: while $K = 2$ still converges, $K = 3$ demands $N \approx 10^{27}$ paths, computationally inconceivable.

Our numerical experiments extend the paper in two directions. First, the worst-case bound is extremely conservative (factor $\sim 10^3$–10^4 for typical targets, Section 3.6). Second, full OLS consistently outperforms quasi-regression at finite N, by up to 70× at $K = 5$.

Our American put experiment confirms that underfitting ($K \leq 4$) produces a persistent bias floor with Laguerre polynomials, while $K \geq 5$ yields proper convergence, the power basis is excluded entirely by its $e^{\mathcal{O}(K^2)}$ conditioning. The practical optimum is $K \in [5, 6]$. More fundamental improvements require either bounded/localised bases or a departure from linear regression altogether [Becker et al., 2019].

Task Division

- **Diane Murzi:** Part I mathematical derivations; correct Delta and Vega estimators including product-rule correction; Python implementation of pathwise estimators; weak convergence plot.
- **Paul Archer:** Part II mathematical framework (Sections 3.2–3.4); Brownian case simulations; Table 3 reproduction; Lemma verification plots.
- **Hamza El Arji:** Part I pathwise Greeks implementation with BB correction; lognormal simulations and Table 4; Gram conditioning and basis-family analysis.
- **Chedi Mnif:** American put pricing experiment (Section 3.7); critical assessment; report editing and LaTeX formatting.

References

S. Becker, P. Cheridito, and A. Jentzen. Deep optimal stopping. *Journal of Machine Learning Research*, 20(1):2712–2736, 2019.

P. Glasserman. *Monte Carlo Methods in Financial Engineering*, volume 53. Springer, 2004.

P. Glasserman and B. Yu. Number of paths versus number of basis functions in American option pricing. *Annals of Applied Probability*, 14(4):2090–2119, 2004.

F. A. Longstaff and E. S. Schwartz. Valuing american options by simulation: a simple least-squares approach. *Review of Financial Studies*, 14(1):113–147, 2001.

J. N. Tsitsiklis and B. Van Roy. Regression methods for pricing complex American-style options. *IEEE Transactions on Neural Networks*, 12:694–703, 2001.